Final Model

Best Performing Model:

- **Stacking Ensemble Model**
- Features: Engineered features (tenure_group, avg_monthly_spend, service_count, engagement_score) + encoded categorical variables

Performance Metrics:

- Accuracy: ~85–87%
- ROC-AUC: ~0.89
- F1-score: ~0.83

Explainability (SHAP):

- Top factors influencing churn:
    - High MonthlyCharges → increases churn risk
    - Low tenure → increases churn risk
    - Few services subscribed → increases churn risk

Customer Segmentation (KMeans):

- 3 Segments identified:
    1. High-Risk (short tenure, fewer services)
    2. Loyal (long tenure, multiple services)
    3. Moderate

Business Recommendations:

- Offer discounts or incentives for high monthly charges
- Improve onboarding for low-tenure customers
- Bundle services for medium tenure customers to increase loyalty

Deployment:

- Streamlit App created for live predictions:
    - Input features → Predict churn → Show recommendation + SHAP explanation

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Conclusion:
This project demonstrates a **full ML lifecycle**: data collection → preprocessing → feature

engineering → model training → evaluation → explainability → business insights → deployment.
It is ready to be shared on **GitHub** with modular code, notebooks, reports, and a Streamlit app for interactive demonstration.